

3(a)(i)	0.95 (no units) ; ;	2
3(a)(ii)	<i>any two from:</i> 1 B, run for less time / stopped early ; 2 chlorophyll, converted / changed, to phaeophytin ; 3 AVP ;	2
3(a)(iii)	<i>any one from:</i> 1 different solvents ; 2 different, TLC plate / chromatogram / stationary phase (material) ; 3 different temperatures ;	1

Question	Answer	Marks
3(b)	any four from: <i>similarities</i> 1 both absorb, blue / between 400 and 475 nm ; 2 both absorb, little / no, light between 525 and, 600 / 630, nm or both absorb, little / no, green / yellow, light ; 3 AVP ; <i>max three differences</i> 4 chlorophyll <i>a</i> absorbs, at more / a larger range of, wavelengths ; 5 chlorophyll <i>a</i> absorbs, more / better, between 400 and 445 nm ; 6 carotenoids absorb, more / better, between 445 and 525 nm ; 7 chlorophyll <i>a</i> absorbs between, 600 / 640, and, 680 / 690 nm, <u>and</u> carotenoids do not ; 8 chlorophyll <i>a</i> (blue absorption) peak / highest, at 425 / 430 nm <u>and</u> carotenoids (blue absorption) peak / highest, at 460 / 465 nm ; 9 chlorophyll <i>a</i> has blue peak and red peak but carotenoids have two blue peaks and no red peak or chlorophyll <i>a</i> has two peaks but carotenoids have three peaks ;	4
3(c)	any four from: 1 gibberellin, moves to / acts on / stimulates, <u>aleurone</u> layer ; 2 binds to, (gibberellin) receptor / GID(1) ; 3 breaks down <u>DELLA</u> (protein) ; 4 releases, PIF / transcription factor ; 5 switches on <u>gene</u> for, amylase / maltase / protease ; 6 amylase, breaks down / hydrolyses, starch to maltose ; 7 sugars / glucose, for, respiration / growth, of embryo ;	4